

## Module 6: Metabolism — Study Questions

### Energy and Thermodynamics

1. What is metabolism? How does it differ from a single chemical reaction?
2. What is the difference between kinetic energy and potential energy? Give a biological example of each.
3. State the First Law of Thermodynamics. Why can't organisms create their own energy from nothing?
4. State the Second Law of Thermodynamics. What is entropy?
5. If entropy always increases, how do living organisms maintain their complex, ordered structures?
6. What is the difference between an exergonic and an endergonic reaction? Which has a negative  $\Delta G$ ?

### Enzymes

1. What is activation energy? Why do reactions need it even if they are exergonic?
2. How do enzymes lower activation energy? Are enzymes consumed in the reaction?
3. What is the active site of an enzyme? How does the induced-fit model differ from the lock-and-key model?
4. What factors affect enzyme activity? Explain how temperature, pH, and substrate concentration each influence reaction rate.
5. What is competitive inhibition? How does it differ from non-competitive (allosteric) inhibition?
6. What happens to an enzyme if it is denatured? Can denaturation be reversed?

7. What is a cofactor? What is a coenzyme? Give an example of each.

## **ATP and Energy Coupling**

1. Describe the structure of ATP. Which bond stores the energy that cells use?
2. How does ATP release energy? Write the reaction.
3. What is energy coupling? Why is it essential for cells to perform endergonic reactions?
4. Approximately how much ATP does a human body use and recycle per day?

## **Metabolic Pathways**

1. What is the difference between anabolic and catabolic pathways? Classify photosynthesis and cellular respiration.
2. What are oxidation-reduction (redox) reactions? In a redox pair, which molecule is oxidized and which is reduced?
3. Explain the mnemonic OIL RIG. How does it help you remember redox reactions?
4. What are  $\text{NAD}^+$  and FAD? What role do these electron carriers play in metabolism?
5. What is feedback inhibition? Draw a simple pathway diagram and show where the end product acts to regulate it.
6. A cell is producing too much of amino acid X. Amino acid X is made through a five-step pathway. Using your knowledge of feedback inhibition, predict how the cell regulates this pathway efficiently.
7. Why is metabolic regulation important? What would happen if all metabolic pathways ran at full speed all the time?